Part IIB Revision Notes

4F7: Statistical Signal and Network Models
Networks

Qianshuo (Harry) Ye
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1 Introduction to Networks

Adjacency matrix A: binary valued and symmetric matrix with

$$A_{ij} = \begin{cases} 1 & \text{if there is an edge from node } i \text{ to node } j \\ 0 & \text{otherwise} \end{cases}$$

Degree of a node k_i: number of edges incident to the node

$$k_i = \sum_{j=0}^{n-1} A_{ij}$$

The average (mean) degree is $k = \frac{1}{n}\sum_{i=0}^{n-1} k_i = \frac{2m}{n}$.

Bipartite graph: a graph where the nodes can be split into 2 groups, such that all edges fall between groups and none within groups.

Components: a maximal set of nodes that can be reached from on another. An undirected graph is *connected* if it has only one component.

Shortest path $s(i,j)$: the minimum number of edges that must be traversed to reach one node i from the other node j. If i and j are directly connected, $s(i,j) = 1$. The shortest path is not defined for nodes in different components. Properties of shortest path length (distance): $s(i,i) = 0$, $s(i,j) = s(j,i)$, and $s(i,j) \leq s(i,k) + s(k,j)$.

Diameter: the longest shortest path between any two nodes in the component.

2 Network Metrics

2.1 Node Importance

Eigenvector centrality: the importance of a node (x_i) is proportional to the sum of the importance of its neighbors.

$$x_i = \alpha \sum_{j=0}^{n-1} A_{ij} x_j$$

For some constant α. In matrix form, $\boldsymbol{x} = \alpha A \boldsymbol{x}$, which means $\boldsymbol{x}$ is an eigenvector of A. If $x_i \geq 0$, the solution is unique and will correspond to the eigenvector of A with the largest eigenvalue.

PageRank: the importance of a node is the probability that a websurfer is on that node in the long time limit. The websurfer randomly clicks an random outgoing link with probability α and jumps to a random node with probability $1-\alpha$.

$$x_i = \alpha \sum_{j=0}^{n-1} \underbrace{\left(\frac{A_{ji}}{k_j}\right)}_{\text{click from } j} x_j + \underbrace{\frac{1-\alpha}{n}}_{\text{random jump}}$$

Considering the probability of going k steps without teleporting

$$x_i = \sum_{k=0}^{\infty} (1-\alpha)\alpha^k \left(\frac{1}{n}\sum_{j=0}^{n-1}(W^k)_{ji}\right) = \frac{1-\alpha}{n}\sum_{j=0}^{n-1}\left((I-\alpha W)^{-1}\right)_{ji}$$

where $W_{ij} = A_{ij}/k_j$ is the walk matrix.

Betweenness centrality: the importance of a node is proportional to the number of shortest paths that pass through it.

$$b_i = \sum_{j,k} \frac{\overbrace{s_{jki}}^{\text{\# shortest paths through } i}}{\underbrace{s_{jk}}_{\text{\# shortest paths between } j \text{ and } k}}$$

For a fully connected graph, $b_i = 0$ for all nodes i. For two separate fully connected graphs, each with n nodes, connected by a single edge, the nodes on either side of the edge have $b_i = (n-1)n$ and all other nodes have $b_i = 0$.

2.2 Clustering Coefficient

A triangle forms when

$$A_{ij}A_{jk}A_{ik} = 1$$

Local clustering coefficient:

$$c_i = \frac{2 \times \text{\# triangles with node } i}{2 \times \text{\# maximum possible triangles with node } i} = \frac{\sum_{j,k} A_{ij}A_{jk}A_{ik}}{k_i(k_i-1)}$$

A clustering coefficient for the whole network is the average of c_i: $c = \frac{1}{n}\sum_i c_i$.

Global clustering coefficient (transitivity):

$$c = \frac{6 \times \text{\# triangles}}{6 \times \text{\# maximum possible triangles}} = \frac{\sum_{i,j,k} A_{ij}A_{jk}A_{ik}}{\sum_i k_i(k_i-1)}$$

2.3 Assortativity and Modularity

Motivation: 'homophily' – nodes with similar properties tend to connect to each other.

Assortativity: a measure of the extent to which nodes' properties are correlated across edges. The assortativity score is the Pearson correlation:

$$r = \frac{\text{cov}(x_i, x_j)}{\sqrt{\text{var}(x_i)\text{var}(x_j)}} = \frac{\frac{1}{2m}\sum_{i,j} A_{ij}x_i x_j \; - \; \left(\frac{1}{2m}\sum_i k_i x_i\right)^2}{\frac{1}{2m}\sum_i k_i x_i^2 \; - \; \left(\frac{1}{2m}\sum_i k_i x_i\right)^2}$$

If $x_i = x_j$ whenever $A_{ij} = 1$, then $r = 1$ and corresponds to *perfect assortativity*. If $x_i = -x_j$ whenever $A_{ij} = 1$, then $r = -1$ and corresponds to *perfect disassortativity*.

Modularity: generalises assortativity for categories, beyond numerical quantities.

Notations: g_i = category of node i, $\delta_{g_i,r} = 1$ if $g_i = r$ and 0 otherwise.

The modularity score quantifies how likely nodes in group r are connect to others in the same group:

$$Q = \sum_r \left(\frac{1}{2m} \sum_{i,j} A_{ij} \underbrace{\delta_{g_i,r}\delta_{g_j,r}}_{\delta_{g_i,g_j}} - \left(\frac{1}{2m} \sum_i k_i \delta_{g_i,r} \right)^2 \right) = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta_{g_i,g_j}$$

3 Spectral Methods and Interpolation

3.1 Linear Interpolation

Interpolate y_i for node i using

$$y_i = \begin{cases} y_i & \text{if node } i \text{ is observed} \\ \frac{1}{k_i}\sum_j A_{ij} y_j & \text{otherwise} \end{cases}$$

Define $W_{ij} = A_{ij}/k_i, y_i = \sum_j W_{ij} y_j = \sum_{j:\text{obs.}} W_{ij} y_j + \sum_{j:\text{unobs.}} W_{ij} y_j = b_i + \sum_{j:\text{unobs.}} W_{ij} y_j$.

Define $W_{ij}^{(u)} = W_{ij}$ for unobserved nodes i and j gives

$$\begin{aligned} \boldsymbol{y} &= \boldsymbol{b} + W^{(u)}\boldsymbol{y} \\ \boldsymbol{y} &= \left(I - W^{(u)}\right)^{-1} \boldsymbol{b} \end{aligned}$$

3.2 Graph Fourier Transform

Laplacian operator $\mathcal{L}$ is defined as:

$$\mathcal{L}f = \frac{\mathrm{d}^2 f}{\mathrm{d}x^2} = \lim_{h \to 0} \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

On a graph, $h = 1$ and $f(x) - f(x+h)$ corresponds to $y_i - y_j$, yielding:

$$(\mathcal{L}\boldsymbol{y})_i = \sum_j A_{ij}(y_i - y_j) = k_i y_i - \sum_j A_{ij} y_j$$

The Laplacian matrix is $\mathcal{L} = D - A$, where D is the diagonal degree matrix with $D_{ii} = k_i$ and $D_{ij} = 0$ for $i \neq j$. The corresponding eigenvalue equation is $\mathcal{L}\boldsymbol{v} = \lambda \boldsymbol{v}$.

The signals can be written as a series in the eigenvectors of $\mathcal{L}$:

$$\boldsymbol{y} = \sum_{\alpha=0}^{n-1} c_\alpha \boldsymbol{v}_\alpha \text{ with } c_\alpha = \boldsymbol{v}_\alpha^\top \boldsymbol{y}$$

Facts:

1. The Laplacian matrix $\mathcal{L}$ is positive semi-definite, i.e. $\lambda_i \geq 0$ for all i.
2. The constant vector $\boldsymbol{v}_0 = (1, 1, \ldots, 1)^\top$ is an eigenvector of $\mathcal{L}$ with eigenvalue $\lambda_0 = 0$.
3. The eigenvalue $\lambda_1 = 0$ if and only if the graph is not connected.

3.3 Spanning Trees

Tree: a network without any cycles.

Spanning tree: a subset of edges that forms a tree on its nodes.

Deletion-contraction relation: the number of spanning trees in a graph G is equal to the number of spanning trees in G with an edge e deleted ($G \backslash e$) plus the number of spanning trees in G with the same edge e contracted (G/e).

$$t(G) = t(G \backslash e) + t(G/e)$$

Facts:

1. The number of spanning trees for a graph with Laplacian $\mathcal{L}$ is equal to $|\mathcal{L}^{(i,i)}|$ (determinant of the matrix obtained by removing the i-th row and column of $\mathcal{L}$), for any i.
2. The number of spanning trees of a graph is $\frac{1}{n} \prod_{i=1}^{n-1} \lambda_i$.

3.4 Normalised Laplacian

Random walk Laplacian: $\mathcal{L}^{(\text{rw})} = I - W$ with $W_{ij} = A_{ij}/k_i$.

Symmetric normalised Laplacian:

$$\mathcal{L}_{ij}^{(\text{sn})} = \begin{cases} 1 & \text{if } i = j \\ -\frac{A_{ij}}{\sqrt{k_i k_j}} & \text{otherwise} \end{cases}$$

Relationship between different Laplacians:

$$\begin{aligned} \mathcal{L} &= D - A \\ \mathcal{L}^{(\text{rw})} &= D^{-1}\mathcal{L} \text{ (average the differences)} \\ \mathcal{L}^{(\text{sn})} &= D^{-\frac{1}{2}}\mathcal{L}D^{-\frac{1}{2}} \text{ (symmetric)} \end{aligned}$$

3.5 Clustering

Find an $\boldsymbol{x}$ that minimises

$$\sum_{i,j} A_{ij}(x_i - x_j)^2 = \boldsymbol{x}^\top \boldsymbol{\mathcal{L}} \boldsymbol{x}$$

and assign group 1 if $x_i \geq 0$ and group 2 if $x_i < 0$, subject to constraints:

1. $\sum_i x_i = 0$ – to avoid nodes to be in one group.
2. $\sum_i x_i^2 = 1$ – to ensure variation.

This yields $\boldsymbol{x}^\top \boldsymbol{\mathcal{L}} \boldsymbol{x} = \sum_{j=1}^{n-1} c_j^2 \lambda_j$, which is minimised when $\boldsymbol{x} = \boldsymbol{v}_1$, the second eigenvector of $\mathcal{L}$.

4 Random Graph Models

The random graph $G(n,p)$: a graph with n nodes where each pair of nodes is connected with probability p. The adjacency matrix follows $A_{ij} \sim \text{Bernoulli}(p)$. For an undirected graph (symmetric adjacency matrix, only consider each edge once)

$$P(A) = \prod_{i<j} p^{A_{ij}} (1-p)^{1-A_{ij}}$$

The probability of a node to be degree k is $p_k = \binom{n-1}{k} p^k (1-p)^{n-1-k}$

Given an adjacency matrix, the maximum likelihood estimate of p is $\hat{p} = \frac{2m}{n(n-1)}$.

Poisson random graph: a random graph with $p = \frac{\lambda}{n}$. With $n \to \infty$, the degree distribution converges to a Poisson distribution $p_k = \frac{\lambda^k e^{-\lambda}}{k!}$.

Giant component: If p were close to 1, as $n \to \infty$, the size of the largest component would go to infinity – the giant component.

Facts:

1. There is at most one giant component in a random graph.
2. For $p > 1/n$, there exists a giant component.

Triangles: In a Poisson random graph, the total number of triangles is

$$\mathbb{E}\left[\sum_{i<j<k} A_{ij} A_{jk} A_{ik}\right] = \binom{n}{3} p^3 \to \frac{\lambda^3}{6} \text{ as } n \to \infty$$

The probability for a randomly chosen node to have a triangle is 0.

Squares: In a Poisson random graph, the total number of cycles of length 4 is

$$\mathbb{E}[\text{\#4-cycles}] = 3\binom{n}{4} p^4 (1-p)^2 \to \frac{\lambda^4}{8} \text{ as } n \to \infty$$

General motifs: For a motif G' with s nodes and k edges, the probability of seeing a G' in a random graph is proportional to $p^k(1-p)^{\binom{s}{2}-k} = \mathcal{O}(n^{-k})$.

$$\mathbb{E}[\#G'] = \binom{n}{s} \underbrace{Z(G')}_{\text{a property of } G'} p^k(1-p)^{\binom{s}{2}-k} = \mathcal{O}(n^{s-k}) \to \begin{cases} 0 & \text{if } k > s \\ \text{const.} & \text{if } k = s \end{cases}$$

Implication: as $n \to \infty$, a randomly chosen node is unlikely to be part of any cycles, hence the random graph is locally tree-like.

The small components: any node not in the unique giant component must be in a small component. The small components are distributed as

$$\pi(s) = \frac{(s\lambda)^{s-1}e^{-s\lambda}}{s!}$$

When $\lambda < 1$, all components are small with high probability. When $\lambda > 1$, there exists a giant component. When $\lambda = 1$, $\pi(s) \approx \frac{1}{\sqrt{2\pi}}s^{-3/2}$ and $\mathbb{E}[s] = \infty$.

Unrealistic properties of the random graph:

1. Locally tree-like and does not contain a high density of short cycles.
2. Has a Poisson degree distribution, rather than a heavy-tailed degree distribution.

4.1 Watts-Strogatz Model

To sample from a Watts-Strogatz model:

1. Create a cycle on n nodes.
2. Connect each node to its k (should be even) nearest neighbors on the cycle. Each node is in $\frac{3k(k-2)}{8}$ traingles and the average shortest path length is $l = \frac{n+k-1}{2k}$.
3. For every edge (i, j), rewire it to (i, j') for a randomly selected node j' with probability p.

Eventually, every node will have at least $k/2$ edges. The number of extra edges is approximately Poisson distributed with mean $k/2$. The large components fo these networks have low diameter.

Fact: For any fixed $p > 0$, the Watts-Strogatz model generates the small-world property, i.e. the distance between nodes is $\mathcal{O}(\log n)$.

4.2 Preferential Attachment Model (Barabási-Albert Model)

In a preferential attachment model, nodes are added to the network one at a time. As a new node arrives, it preferentially attaches to c nodes with a large number of existing edges, i.e. proportional to the degree.

Given n nodes in the network, the $(n+1)$th node selects node i to connect to with probability

$$\frac{k_i}{\sum_j k_j} = \frac{k_i}{2cn}$$

Fact: The preferential model has a power-law degree distribution $p_k \sim k^{-3}$.

Note:

- The distribution is heavy-tailed
- Mean degree $\sum_k kp_k = 2c$
- the variance of degree distribution is $\sum_k (k-2c)^2 p_k = \infty$ as $n \to \infty$

4.3 Graphs with Arbitrary Degree Distribution

Configuration model: For each node i, assign it a degree k_i drawn from the degree distribution p_k. Create k_i stubs (half-edges) and connect stubs at random to form edges. The resultant graph is a multigraph with exactly the degree sequence $\{k_i\}$.

Excess degree distribution q_k: probability that a node reached by following a random edge has k other edges. This is proportional to the number of edges that can be followed to reach the node, i.e. $q_k \propto (k+1)p_{k+1}$, normalised by $1/\mathbb{E}[k]$.

The generating function for q_k is

$$g_q(z) = \frac{g_p'(z)}{g_p'(1)} = \frac{\sum_k kp_k z^{k-1}}{\mathbb{E}[k]} = \sum_{k=0}^{\infty} q_k z^k$$

Define u = probability a random not is not in the giant component, and w = probability a random edge does not lead to the giant component, then u and w satisfy:

$$u = \sum_k p_k w^k = g_p(w)$$
$$w = \sum_k q_k w^k = g_q(w) = \frac{g_p'(w)}{g_p'(1)}$$

Fact: A random network with degree distribution p_k will have a giant component if $\mathbb{E}[k^2] > 2\mathbb{E}[k]$.

4.4 Graphs with Community Structure

Stochastic block model: a model that accounts for varying densities within and between groups. Each node i is placed in one of K groups at random, i.e. $g_i \in \{1, \ldots, K\}$. The adjacency matrix is generatated from $A_{ij} \sim$ Bernoulli($\omega_{g_i g_j}$), where $\omega_{g_i g_j}$ is the probability of an edge between nodes in groups g_i and g_j. With group assignment $\boldsymbol{g}$, the probability for an adjacency matrix is

$$P(A|\boldsymbol{g}, \omega) = \prod_{i<j} \omega_{g_i g_j}^{A_{ij}} (1 - \omega_{g_i g_j})^{1-A_{ij}}$$

5 Community Detection

Goal: find statistically related nodes in a network.

5.1 Clustering

To recover the communities $\boldsymbol{g}$ given a network A, apply Bayes' law:

$$P(\boldsymbol{g}|A,\omega) = \frac{P(A|\boldsymbol{g},\omega)P(\boldsymbol{g})}{P(A|\omega)} = \frac{1}{Z}\prod_{i<j}\omega_{g_ig_j}^{A_{ij}}(1-\omega_{g_ig_j})^{1-A_{ij}}$$

Compute the posterior probability Q_{ir} that node i is in group r using naive Bayes:

$$\begin{aligned} Q_{ir} = P(g_i = r|A,\omega) &= \sum_{\boldsymbol{g}\setminus i} P(g_i = r, \boldsymbol{g}\setminus_i|A,\omega) \\ &\propto \sum_{\boldsymbol{g}\setminus i}\left(\prod_j \omega_{rg_j}^{A_{ij}}(1-\omega_{rg_j})^{1-A_{ij}}\right)\underbrace{\left(\prod_{j<k,\neq i}\omega_{g_jg_k}^{A_{jk}}(1-\omega_{g_jg_k})^{1-A_{jk}}\right)}_{\text{posterior when } i \text{ is removed}} \end{aligned}$$

Assuming (1) removing a single node i would not change the posterior for the rest and (2) the posterior term factorises (true in appropriate limits) gives:

$$\begin{aligned} Q_{ir} &\propto \sum_{\boldsymbol{g}\setminus i}\prod_j Q_{jg_j}\omega_{rg_j}^{A_{ij}}(1-\omega_{rg_j})^{1-A_{ij}} \\ &= \prod_j\sum_s \omega_{rs}^{A_{ij}}(1-\omega_{rs})^{1-A_{ij}}Q_{js} \end{aligned}$$

Given Q_{ir}, the minimum error is achieved by setting $\hat{g}_i = \arg\max_r Q_{ir}$.

5.2 Parameter Estimation

Find ω that maximises the likelihood of the observed network A:

$$\begin{aligned} \log P(A|\omega) &= \log\sum_{\boldsymbol{g}} P(A,\boldsymbol{g}|\omega) \\ &\geq \sum_{\boldsymbol{g}} Q(\boldsymbol{g})\log\frac{P(A,\boldsymbol{g}|\omega)}{Q(\boldsymbol{g})} \quad \text{(Jensen's inequality)} \end{aligned}$$

Setting $Q(\boldsymbol{g}) = P(\boldsymbol{g}|A,\omega)$ equates the LHS and the RHS, and the optimal ω is

$$w_{rs} = \frac{\sum_{i,j} A_{ij}Q_{ir}Q_{js}}{\sum_{i,j} Q_{ir}Q_{js}}$$

Use EM algorithm to iteratively update Q_{ir} and ω until convergence:

1. E-step: update Q_{ir} using the current ω.
2. M-step: update ω using the current Q_{ir}.